

Quiz 1

Instructions: You must show all of your work to receive credits for correct answers.**Problem 1.** (4 points) Evaluate the following integrals

(a) $\int x^2 + \sqrt[3]{x} \, dx$

$$= \int x^2 + x^{\frac{1}{3}} \, dx$$

$$= \frac{x^3}{3} + \frac{3}{4}x^{\frac{4}{3}} + C$$

(b) $\int_0^{\ln 2} e^{2x} \, dx$

$$= \frac{1}{2} [e^{2x}]_{x=0}^{x=\ln 2}$$

$$= \frac{1}{2} [e^{2\ln 2} - e^{2 \cdot 0}]$$

$$= \frac{1}{2} [e^{\ln 4} - e^0] = \frac{1}{2} (4 - 1) = \frac{3}{2}$$

Problem 2. (6 points) Evaluate the following integrals using appropriate substitutions:

(a) $\int \frac{2x}{x^2+2} \, dx$

let $u = x^2 + 2$, $du = 2x \, dx$.

$$\int \frac{2x}{x^2+2} \, dx \xrightarrow{u=x^2+2} \int \frac{1}{u} \, du$$

$$= \ln|u| + C$$

$$= \ln(x^2+2) + C$$

(b) $\int \cos \theta \sin^4 \theta \, d\theta$

let $u = \sin \theta$, $du = \cos \theta \, d\theta$.

$$\int \cos \theta \sin^4 \theta \, d\theta$$

$$\xrightarrow{u=\sin \theta} \int u^4 \, du$$

$$= \frac{1}{5} u^5 + C = \frac{1}{5} \sin^5 \theta + C$$

(c) $\int_1^e \frac{(\ln x)^2}{x} \, dx$

let $u = \ln x$, $du = \frac{1}{x} \, dx$

when $x=1$, $u=\ln 1=0$

$x=e$, $u=\ln e=1$

$$\int_1^e \frac{(\ln x)^2}{x} \, dx \xrightarrow{u=\ln x} \int_0^1 u^2 \, du$$

$$= \left[\frac{u^3}{3} \right]_{u=0}^{u=1} = \frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3}$$

(d) $\int_0^1 t e^{-t^2} \, dt$

let $u = -t^2$, $du = -2t \, dt$

when $t=0$, $u=-0^2=0$

$t=1$, $u=-1^2=-1$

$$\int_0^1 t e^{-t^2} \, dt \xrightarrow{u=-t^2} \int_0^{-1} e^u \left(-\frac{1}{2} du\right)$$

$$= -\frac{1}{2} [e^u]_{u=0}^{u=-1} = -\frac{1}{2} (e^{-1} - e^0)$$

$$= -\frac{1}{2} \left(\frac{1}{e} - 1\right) = \frac{1}{2} - \frac{1}{2e}$$